AQA

A-Level
Economics

7136/1 Paper 1 Markets And Market Failure
Predicted Paper Final Mark scheme

7136
Predicted Paper

Version/Stage: v1.0   [Set C — Advanced]

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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This is a predicted paper intended for revision purposes only. It is not an official AQA publication. The mark scheme follows the style and structure of AQA published materials to support realistic practice.

Annotations

Annotation	Description
?	Unclear
AN	Analysis
APP	Application
BOD	Benefit of the doubt
EVAL	Evaluation
Highlight	Highlight
IR	Irrelevant
KU	Knowledge and understanding
NAQ	Not answered question
REP	Repetition
SEEN	Indicates that the point has been noted, but no credit has been given.
Tick	Correct point
Cross	Incorrect point

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 — Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best-fit approach for defining the level and then use the variability of the response to help decide the mark within the level.

Step 2 — Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Levels of response marking grid — 25-mark questions

The following generic grid should be used when marking all 25-mark questions (questions 0 4, 0 8, 1 0, 1 2 and 1 4 on paper 7136/1).

Level of response	Response	Max 25 marks
5	Sound, focused analysis and well-supported evaluation that: • is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response • includes well-focused analysis with clear, logical chains of reasoning • includes supported evaluation throughout the response and in a final conclusion.	21–25 marks
4	Sound, focused analysis and some supported evaluation that: • is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes some good application of relevant economic principles to the given context and, where appropriate, some good use of data to support the response • includes some well-focused analysis with clear, logical chains of reasoning • includes some reasonable, supported evaluation.	16–20 marks
3	Some reasonable analysis but generally unsupported evaluation that: • focuses on issues that are relevant to the question, showing satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response • includes some reasonable analysis but which might not be adequately developed or becomes confused in places • includes fairly superficial evaluation; there is likely to be some attempt to make relevant judgements but these aren't well-supported by arguments and/or data.	11–15 marks
2	A fairly weak response with some understanding that: • includes some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • includes some limited application of relevant economic principles to the given context and/or data to the question • includes some limited analysis but it may lack focus and/or become confused • includes some evaluation which is weak and unsupported.	6–10 marks
1	A very weak response that: • includes little relevant knowledge and understanding of economic terminology, concepts and principles • includes application to the given context which is, at best, very weak • includes attempted analysis which is weak and unsupported.	1–5 marks

Section A

Context 1 **Total for this Context: 40 marks**

The UK National Living Wage and labour market inequality

0 1 Using the data in Extract A (Figure 1), calculate the percentage change in real median weekly earnings between 2014 and 2024. **[2 marks]**

Calculation:

(578 – 542) / 542 × 100 = 36/542 × 100 = **6.6%** (increase)

Response	Max 2 marks
For the correct answer (6.6% (increase)) with correct working shown.	2 marks
For correct working with an arithmetic slip; OR correct numerical answer without working shown.	1 mark

0 2 Explain how the data in Extract A (Figure 2) show that the effects of the National Living Wage are concentrated in particular sectors of the UK economy. **[4 marks]**

Response	Max 4 marks
• includes evidence that clearly shows that the effects of the National Living Wage are concentrated in particular sectors. • clearly explains how this data is evidence that the effects of the National Living Wage are concentrated in particular sectors.	4 marks
• includes evidence that shows that the effects of the National Living Wage are concentrated in particular sectors. • unclear explanation of how this data is evidence that the effects of the National Living Wage are concentrated in particular sectors.	3 marks
• includes evidence that shows that the effects of the National Living Wage are concentrated in particular sectors. • limited or no explanation of how this data is evidence that the effects of the National Living Wage are concentrated in particular sect	2 marks
• includes evidence that does not clearly show that the effects of the National Living Wage are concentrated in particular sectors. • weak or no explanation.	1 mark

Relevant issues include:

- explanation of the concept of sectoral concentration of NLW impact
- low-paid sectors have very high shares of NLW workers: hospitality (28.3%), retail (22.7%), social care (19.1%)
- professional services has only 0.4% NLW workers — near-zero incidence — indicating the NLW has virtually no direct effect on this sector
- median hourly wage in hospitality (£11.90) is very close to the NLW rate (£11.44) — meaning a large proportion of workers are at or close to the floor
- median hourly wage in professional services (£22.85) is nearly double the NLW, confirming NLW policy does not directly bind in this sector
- implication: employment and pay effects of NLW changes will be highly sector-specific, concentrated in hospitality, retail and social care

0 3 Extract B states that 'a well-targeted minimum wage can then raise pay without reducing employment.' With the help of a monopsony diagram, explain how the introduction of a minimum wage may raise both wages and employment in a monopsonistic labour market. **[9 marks]**

Levels of response marking grid — 9-mark questions

The following generic grid should be used when marking all 9-mark questions in Section A (questions 0 3 and 0 7).

Level of response	Response	Max 9 marks
3	• is well organised and develops one or more of the key issues that are relevant to the question • shows sound knowledge and understanding of relevant economic terminology, concepts and principles • includes good application of relevant economic principles and/or good use of data to support the response • includes well-focused analysis with a clear, logical chain of reasoning • includes a relevant diagram that will, at the top of this level, be accurate and used appropriately.	7–9 marks
2	• includes one or more issues that are relevant to the question • shows reasonable knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles and/or data to the question • includes some reasonable analysis but it might not be adequately developed and may be confused in places • may include a relevant diagram.	4–6 marks
1	• is very brief and/or lacks coherence • shows some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • demonstrates very limited ability to apply relevant economic principles and/or data to the question • may include some very limited analysis but the analysis lacks focus and/or becomes confused • may include a relevant diagram but the diagram is not used and/or is inaccurate in some respects.	1–3 marks

Suggested diagram:

A monopsony labour market diagram is expected. Axes: wage rate / quantity of labour. Upward-sloping labour supply curve (S = ACL — the average cost of labour the monopsonist faces). The marginal cost of labour (MCL) curve lies above and steeper than S, because hiring an extra worker requires raising wages for all employees. Downward-sloping MRPL (= demand for labour). The profit-maximising monopsonist hires at MCL = MRPL, giving quantity L1, then pays the wage at which L1 workers are willing to work (read from S), giving wage W1. Because the monopsonist exploits its wage-setting power, W1 lies below the competitive-market equilibrium. Introduction of a minimum wage (NMW) above W1 but below the MRPL at L1 makes the labour supply curve horizontal at NMW up to L*, then rising along S. Effectively, MCL = NMW over that range. The firm now employs where NMW = MRPL, at L2 > L1, paying wage NMW > W1. Wages and employment both rise.

Relevant issues include:

- definition of monopsony — a single or dominant buyer of labour facing upward-sloping labour supply
- distinction between MCL (marginal cost of labour) and ACL (average cost of labour), with MCL above ACL because extra hires require wage rises for all workers
- monopsony equilibrium at L1, W1, where MCL = MRPL, with W1 < MRPL — workers paid below their marginal productivity

• sources of monopsony power in the UK: concentrated employment in social care (NHS + a few large chains), retail and hospitality (large chains as dominant local employers)

• mechanics of NMW introduction: horizontal supply segment at NMW flattens the MCL locally

• firm re-optimises: employs up to where NMW = MRPL (for wage up to the competitive level)

• result: both wages AND employment rise, unlike the classical labour-market prediction

• this explains why empirical evidence (e.g. UK NMW since 1999) has consistently failed to find the large disemployment effects predicted by simple competitive models

• other factors: efficiency-wage effects, reduced turnover, possible absorption through lower profits or small price rises

0 4 Extract C states that 'the NLW has already been extended further than evidence can support, and... additional increases risk significant disemployment effects, particularly among younger workers in small firms.' Use the extracts and your knowledge of economics to evaluate the view that further increases in the UK National Living Wage would be the most effective policy for reducing income inequality and in-work poverty. **[25 marks]**

Areas for discussion include:

- UK NLW history and recent changes: from £6.50 (2014) to £11.44 (2024), with extension to age 21+; projected to reach two-thirds of median earnings by 2025
- the case for further increases: directly raises pay for lowest-paid workers; monopsony conditions in key sectors limit disemployment effects (Extract B); evidence of spill-over effects compressing the wage distribution beyond directly-affected workers
- evidence on the UK record: NLW has raised pay with smaller-than-predicted employment effects; 2.7m workers directly affected in 2024 (Extract B)
- the case against further increases: approaching the territory where classical labour-market theory predicts larger employment effects; particular risk for young workers in small firms (Extract C); risk of sectoral concentration of losses (hospitality, retail)
- importantly, the NLW cannot reach households without an earner (pensioners, disabled, unemployed) — so it is not a complete poverty tool
- inequality evidence: despite NLW rising, Gini coefficient has drifted upward (0.340 → 0.363 over 2014–2024) because top-decile incomes have grown faster — NLW addresses the bottom of the distribution but cannot reduce overall inequality
- wealth inequality is not affected by the NLW — top 10% own 43% of wealth; bottom 50% own less than 9%
- alternative / complementary policies: progressive taxation (tackles top of distribution); Universal Credit / in-work benefits (tackles households with low hours); education and skills investment (tackles underlying productivity differentials); regional policy (tackles geographic concentration of high-paid employment); housing policy (tackles cost-of-living element of poverty); union rights
- behavioural economics: present bias means pay rises may not translate fully into long-term welfare gains without complementary measures (e.g. pensions auto-enrolment)
- structural causes: UK productivity weakness, limited vocational training, regional imbalance
- distributional pass-through: who bears the cost of NLW rises? Consumers (via higher prices), profits, workers (via reduced hours), firms (via automation)?

Evaluation / judgement:

- the NLW is effective at the bottom of the wage distribution but cannot alone reduce overall income or wealth inequality
- further increases may offer diminishing returns as the NLW approaches the level at which disemployment effects become material
- a policy mix is usually most effective: NLW + progressive taxation + in-work benefits + skills investment + housing reform
- "most effective" depends on the specific inequality being addressed — for the bottom of the distribution and in-work poverty, the NLW is central; for broader inequality, other instruments are needed
- strongest answers distinguish between types of inequality and poverty, and reach a supported judgement based on evidence

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response to the question.

Use the levels mark scheme for 25-mark questions to award candidates marks for this question.

Section A

Context 2 **Total for this Context: 40 marks**

Market power in digital platform markets

0 5 Using the data in Extract D (Figure 3), calculate, to one decimal place, the difference in the two-firm concentration ratio between the search advertising market and the food delivery market. **[2 marks]**

Calculation:

Search advertising CR2 = 92.8 + 3.5 = 96.3%.

Food delivery CR2 = 34.0 + 33.5 = 67.5%.

Difference = 96.3 – 67.5 = **28.8 percentage points**

Response	Max 2 marks
For the correct answer (28.8 percentage points (or 28.8pp)) with correct working shown.	2 marks
For correct working with an arithmetic slip; OR correct numerical answer without working shown.	1 mark

0 6 Explain how the data in Extract D (Figures 3 and 4) suggest that firms with dominant positions in digital markets are able to earn supernormal profits. **[4 marks]**

Response	Max 4 marks
• includes evidence that clearly shows that firms with dominant positions in digital markets are able to earn supernormal profits. • clearly explains how this data is evidence that firms with dominant positions in digital markets are able to earn supernormal profits.	4 marks
• includes evidence that shows that firms with dominant positions in digital markets are able to earn supernormal profits. • unclear explanation of how this data is evidence that firms with dominant positions in digital markets are able to earn supernormal p	3 marks
• includes evidence that shows that firms with dominant positions in digital markets are able to earn supernormal profits. • limited or no explanation of how this data is evidence that firms with dominant positions in digital markets are able to earn supernorr	2 marks
• includes evidence that does not clearly show that firms with dominant positions in digital markets are able to earn supernormal profi • weak or no explanation.	1 mark

Relevant issues include:

- explanation of the concept of supernormal (abnormal) profit — profit above the minimum required to keep the firm in the industry
- operating margins of dominant tech firms are strikingly high: Microsoft 41.8%, Meta 34.7%, Apple 29.8%, Alphabet 27.4%
- these are far above the typical margins in competitive industries (retail c. 3–5%)
- dominance evidence: Google 92.8% of UK search advertising; Apple iOS app store a 100% walled garden; Microsoft, Apple, Alphabet all have substantial market power in their primary markets
- high margins persist over time, indicating barriers to entry prevent the competitive erosion of supernormal profits that economic theory would predict in a contestable market
- Amazon (6.4%) is a counter-example within the data — lower margins reflect more competitive retail segment; demonstrates that not all tech firms earn supernormal profits equally

0 7 Extract E states that 'These dynamics produce a natural tendency toward market concentration, as smaller rivals struggle to offer equivalent network benefits.' With the help of a diagram, explain how network effects in digital platform markets may lead to monopoly outcomes and a misallocation of resources. **[9 marks]**

Suggested diagram:

A natural-monopoly diagram is a valid approach. Axes: price / quantity. Downward-sloping AR (demand) curve. Long-run AC curve falls continuously over the relevant output range, reflecting increasing returns driven by network effects (each new user adds value to all existing users). MC curve lies below AC throughout. The profit-maximising monopolist equates MC = MR, producing Qm at price Pm, earning supernormal profits (Pm – AC at Qm). The allocatively efficient quantity where P = MC is at Qa > Qm, giving a welfare loss triangle. Alternatively, a standard monopoly diagram showing a natural monopoly in a concentrated market. Either approach acceptable. Network-effects-specific diagram may also show utility of the platform as an upward-sloping function of number of users.

Relevant issues include:

- definition of network effects — the value of a good or service to each user increases as more users adopt it
- direct network effects (social networks, messaging) vs indirect (two-sided platforms — drivers/riders, buyers/sellers)
- self-reinforcing dynamic: more users → more value → more users; leads to "winner-takes-most" outcomes
- data as a barrier to entry: scale generates more data which improves services, which attracts more users
- high fixed costs, low marginal costs — characteristic of digital platforms — amplify scale advantages
- market outcome: monopoly or tight oligopoly, with supernormal profits (evidenced by the operating margins in Fig. 4)
- misallocation: Pm > MC, allocative inefficiency, welfare loss triangle
- distortion of consumer behaviour: lack of alternatives; high switching costs; winner-takes-most means smaller rivals cannot provide equivalent network benefits even if technically superior
- data privacy concerns: consumers may accept terms they would reject in a competitive market
- innovation trade-off: dominant firms may have resources for R&D; but less competitive pressure to innovate (Arrow's replacement effect)

0 8 Extract F states that 'ex-ante regulation in digital markets risks government failure and... ex-post competition enforcement remains the better approach.' Use the extracts and your knowledge of economics to evaluate the view that competition authorities should actively regulate digital platforms rather than relying on the threat of new entry to discipline market power. **[25 marks]**

Areas for discussion include:

- the case for active regulation: network effects create durable monopoly power (Extract E); operating margins of 27–42% (Fig. 4) are consistent with supernormal profits; consumer welfare harms — data exploitation, self-preferencing, killer acquisitions, steering practices
- evidence of market power: Google 92.8% of UK search advertising, Apple iOS 100% App Store share, Uber 56.4% ride-hailing — persistent, not transient
- policy tools: UK Digital Markets, Competition and Consumers Act (2023), EU Digital Markets Act, Strategic Market Status designation, interoperability requirements, data-portability rules, self-preferencing bans
- the case for relying on contestability: MySpace → Facebook, BlackBerry → Apple, rise of TikTok — dynamic competition can displace even entrenched incumbents
- the case against: these transitions took many years, and users suffered consumer surplus losses in the meantime; winners-take-most outcomes are rarely displaced in two-sided markets with strong network effects
- the case for ex-post over ex-ante: flexibility, evidence-based intervention, lower regulatory burden; the CMA's history of intervention (merger blocks, market investigations) provides a precedent
- the case for ex-ante: ex-post enforcement takes years and often comes too late for displaced rivals; deterrent effect of clear rules
- behavioural factors: zero-price pricing obscures consumer welfare analysis; "free" services are paid for in attention and data, which traditional welfare analysis struggles to capture
- government failure risks: regulatory capture given resources of Big Tech; lobbying; unintended consequences (e.g. EU's GDPR entrenching incumbents at the expense of smaller rivals)
- innovation concerns: heavy regulation may chill R&D; investment (evident from Fig. 4 figures — tech firms reinvest 8–28% of revenue in R&D;)
- two-sided markets complicate welfare analysis: predatory pricing against consumers may subsidise innovation on the other side of the market
- international dimension: UK/EU regulators face global firms; risk of fragmentation, risk of under-regulation if jurisdiction shops

Evaluation / judgement:

- the choice is not binary — regulators increasingly combine ex-ante rules (for clearly-designated dominant firms) with ex-post enforcement (for individual abuses)
- the appropriate balance depends on: severity and persistence of market power, effectiveness of alternative instruments, institutional capacity of competition authorities
- evidence suggests pure reliance on entry is inadequate for markets with strong network effects and data-driven barriers
- but heavy-handed ex-ante regulation risks government failure
- strongest answers reach a reasoned judgement recognising the complementarity of the two approaches and the specific features of digital markets

Use the levels mark scheme for 25-mark questions to award candidates marks for this question.

Section B

Levels of response marking grid — 15-mark questions

The following generic grid should be used when marking all 15-mark questions in Section B (questions 0 9, 1 1 and 1 3).

Level of response	Response	Max 15 marks
3	A good response that: • is well organised and develops a selection of the key issues that are relevant to the question • shows sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response • includes well-focused analysis with clear, logical chains of reasoning.	11–15 marks
2	A reasonable response that: • focuses on issues that are relevant to the question • shows satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response • includes some reasonable analysis but which might not be adequately developed or becomes confused in places.	6–10 marks
1	A weak response that: • has identified one or more relevant issues • has some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • has very limited application of relevant economic principles to the given context and/or data to the question • might have some limited analysis but it may lack focus and/or become confused.	1–5 marks

Essay 1 — Total for this Essay: 40 marks

0 9 Explain, with the aid of diagrams, how price and output are determined for a firm in a monopolistically competitive market in both the short run and the long run. **[15 marks]**

Areas for discussion include:

- features of monopolistic competition: many firms, differentiated products, low barriers to entry and exit, some price-making power, good information
- short-run diagram: downward-sloping AR (firm's demand) and MR below it; U-shaped AC; profit-maximising output where MC = MR, giving quantity Q(sr) and price P(sr) > AC, generating supernormal profits (shaded rectangle)
- the supernormal profit attracts new entrants (low barriers to entry)
- long-run diagram: entry shifts each incumbent's AR curve leftward and makes it more elastic (consumers have more substitutes)
- long-run equilibrium where the AR curve is tangent to the AC curve at the profit-maximising output (MC = MR still); P(lr) = AC so only normal profit is earned
- excess capacity result: P > MC, so allocative inefficiency persists; firms operate below the output at which AC is minimised, so productive inefficiency also persists
- trade-off: consumers enjoy variety (differentiated products) at the cost of higher prices than perfect competition would deliver

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate's response to the question.

Use the levels mark scheme for 15-mark questions to award candidates marks for this question.

1 0 Evaluate the view that the degree of contestability of a market is a more important determinant of economic efficiency than the number of firms operating in it. **[25 marks]**

Areas for discussion include:

- definition of contestable markets (Baumol): markets where entry and exit are costless, so potential entrants discipline incumbent behaviour regardless of current number of firms
- conditions for contestability: low sunk costs, absence of legal entry barriers, equal access to technology and inputs; hit-and-run entry feasible
- in perfectly contestable market, even a monopolist would price at zero-profit level and produce at the efficient output
- the case for prioritising contestability: allows economies of scale (natural monopoly) while maintaining pricing discipline; encourages dynamic efficiency through potential entry; less reliant on heavy-handed regulation
- the case against prioritising contestability: pure contestability is rare in practice; network effects, sunk costs, data advantages, brand loyalty all limit contestability; concentration still matters because few real markets are fully contestable
- evidence from UK: budget airlines (Ryanair, easyJet) entered and disciplined incumbents despite high capital costs; Aldi/Lidl have disciplined UK grocery incumbents; tech markets (MySpace → Facebook) show transitions are possible but slow
- counter-evidence: digital platforms (Google search, iOS app store) show durable concentration despite theoretical contestability
- efficiency dimensions: allocative (P = MC), productive (lowest cost), dynamic (innovation) — contestability affects all three via entry threat
- X-inefficiency: Leibenstein's idea that monopolies lose productive efficiency due to absent competitive pressure — contestability may be sufficient to discipline X-inefficiency
- role of competition policy: CMA's approach increasingly focuses on contestability (removing barriers to entry) rather than number of firms
- alternative: some markets are structurally uncontestable (natural monopolies) and require regulation irrespective of contestability arguments

Evaluation / judgement:

- contestability is a powerful concept but its real-world application is often limited by structural barriers (network effects, sunk costs, data)
- "more important" is a relative claim — both contestability and number of firms matter, and their relative importance depends on market structure
- for markets with high structural barriers to entry (digital platforms, natural monopolies), number of firms and structural intervention matter more
- for markets with low structural barriers (retail, hospitality, professional services), contestability may dominate
- strongest answers distinguish types of markets and reach a reasoned judgement

Use the levels mark scheme for 25-mark questions to award candidates marks for this question.

Essay 2 — Total for this Essay: 40 marks

1 1 Explain how insights from behavioural economics, including bounded rationality and present bias, can help to explain market failure in markets such as those for pensions, energy and unhealthy food. **[15 marks]**

Areas for discussion include:

- challenge to the rational-consumer assumption of classical economics
- bounded rationality: consumers lack the information, time, or cognitive capacity to make optimal decisions
- present bias (hyperbolic discounting): consumers systematically over-weight immediate rewards and under-weight future costs
- loss aversion: losses loom larger than equivalent gains — consumers may avoid beneficial switching (e.g. energy tariffs) because of perceived switching costs
- anchoring and framing: consumer decisions are influenced by reference points and how choices are presented
- default effects: consumers often accept default options rather than making active choices
- pensions market: present bias means consumers under-save for retirement; auto-enrolment (UK, 2012) used defaults to address this, with substantial uptake gains
- energy market: information overload and complexity in tariff structures; status-quo bias prevents switching; loyalty penalties exploit inattention
- unhealthy food: present bias over-weights immediate taste satisfaction relative to long-run health costs; bounded self-control makes committing to healthy choices difficult
- result: persistent patterns of behaviour that classical welfare economics treats as revealed preference, but behavioural economics recognises as sub-optimal
- failure goes beyond information asymmetry — even fully-informed consumers exhibit systematic biases

Use the levels mark scheme for 15-mark questions to award candidates marks for this question.

1 2 Evaluate the view that behavioural nudges are a more efficient and equitable method of correcting market failure than taxation or regulation. **[25 marks]**

Areas for discussion include:

- definition of nudges (Thaler and Sunstein): changes to choice architecture that predictably alter behaviour without restricting options or significantly changing incentives
- UK Behavioural Insights Team examples: pensions auto-enrolment (dramatic increase in pension contributions); tax-reminder letters (higher tax compliance); organ donation opt-out
- efficiency case for nudges: low fiscal cost, preserves consumer choice, targeted at specific biases, politically less controversial than taxation
- equity case for nudges: no direct price effect means no regressive impact; available to all consumers equally
- the case for taxation: captures externalities directly via price signal; generates revenue; sugar tax (SDIL) achieved 35% reduction in sugar intake through a combination of consumer and producer response; congestion charging and carbon taxation similarly effective
- the case against nudges alone: may be insufficient for large market failures (e.g. carbon emissions); information-based nudges often fail to reach the groups most at risk (e.g. low financial literacy); voluntary approaches without price signal may not drive deep behaviour change
- the case for regulation: addresses problems where price signals and nudges are ineffective (e.g. product safety, pollution thresholds); clearer outcomes
- the case against regulation alone: can be heavy-handed, inflexible, and create government-failure risks
- equity trade-offs: taxation (especially indirect) is often regressive; regulation may raise costs passed to consumers; nudges may be progressive but often less effective
- behavioural economics insights suggest that the effectiveness of taxation depends on framing, transparency, and consumer attention — nudges and taxes are often complements, not substitutes

- paternalism concerns: nudges have been criticised as "libertarian paternalism" — respecting autonomy in name but manipulating choice architecture in practice
- evidence of a policy mix working well: auto-enrolment + tax relief on pensions (complements); sugar tax + reformulation targets + advertising restrictions (complements)

Evaluation / judgement:

- nudges are often more efficient for small, behavioural problems; taxation and regulation are more effective for large externalities
- "more efficient AND equitable" is a strong claim — often trade-offs exist (e.g. a well-targeted tax may be more effective but more regressive)
- the best policy mix typically combines instruments: nudges to address bias, taxation to price externalities, regulation to set safety floors
- strongest answers recognise the complementarity of the instruments and reach a reasoned judgement based on market specifics

Use the levels mark scheme for 25-mark questions to award candidates marks for this question.

Essay 3 — Total for this Essay: 40 marks

1 3 Explain, with the aid of a diagram, how negative production externalities from carbon emissions lead to an inefficient allocation of resources in a free market. **[15 marks]**

Areas for discussion include:

- definitions: externality, marginal private cost (MPC), marginal social cost (MSC), allocative efficiency, welfare loss
- diagram: upward-sloping MPC (firm's internal cost); MSC above MPC reflecting external costs of carbon emissions; downward-sloping MPB = MSB (assumed no externalities in consumption)
- free-market equilibrium at MPC = MSB, giving Q1
- socially optimal output at MSC = MSB, giving Q*, with Q* < Q1
- welfare loss triangle between Q* and Q1, bounded by MSC and MSB
- over-production and over-consumption relative to social optimum
- scale of the externality: climate change generates global, long-run, irreversible costs
- free-rider problem across nations — individual countries have incentives to under-act
- examples: fossil-fuel power generation, cement and steel manufacturing, aviation and shipping

Use the levels mark scheme for 15-mark questions to award candidates marks for this question.

1 4 Evaluate the view that carbon taxation is the most effective way to address the market failure arising from greenhouse gas emissions in the UK. **[25 marks]**

Areas for discussion include:

- case for carbon taxation (Pigouvian tax): internalises the externality at the source; sets a clear price signal; revenue-raising potential; technology-neutral (lets the market find lowest-cost abatement routes)
- UK examples: Climate Change Levy, Carbon Price Support, fuel duty
- diagrammatic analysis: tax equal to marginal external cost shifts MPC to MSC, achieving Q* = socially optimal output
- the case for cap-and-trade instead: UK ETS sets quantity directly and allows the market to determine the carbon price; provides more certainty about emissions; EU experience shows it can be effective
- the case for regulation: emission standards, energy-efficiency requirements, bans on internal-combustion engine sales (UK by 2035) — clearer outcomes, easier to enforce
- the case for subsidies: support for renewable energy (Contracts for Difference, feed-in-tariffs), electric vehicle subsidies, heat-pump grants — but subsidies are revenue-negative and may pay for already-planned investment
- the case for information-based approaches: carbon labelling, energy-performance certificates for buildings — low-cost but limited effect
- the case for behavioural nudges: default energy tariffs, smart-meter presentation — helpful at margin but insufficient for systemic change
- distributional impact: carbon tax is regressive without recycling of revenue to lower-income households (e.g. carbon dividend, targeted transfers)
- effectiveness depends on price elasticity of demand: if short-run demand is inelastic (e.g. domestic heating), tax raises revenue without rapid quantity response; long-run effects can be larger as capital stock adjusts
- international dimension: unilateral UK action can cause carbon leakage (industry moves overseas); Carbon Border Adjustment Mechanism addresses this
- government failure risks: political pressure leading to exemptions (farming, aviation); revenue earmarking distortions; regulatory capture
- dynamic efficiency: carbon pricing encourages R&D; into clean technologies, delivering long-run benefits beyond the static externality correction
- the overall UK policy mix: ETS for industry + fuel duties + renewable subsidies + regulation of vehicles and buildings + R&D; support — each tackles different failures

Evaluation / judgement:

- carbon taxation is a powerful instrument and likely the backbone of efficient climate policy — but not sufficient alone
- "most effective" depends on criteria: pure allocative efficiency (tax wins), certainty of emissions outcome (cap-and-trade wins), fairness (combination with redistribution wins), political feasibility (regulation and subsidy often wins)
- a comprehensive policy mix outperforms any single instrument — and this is how real-world climate policy is actually implemented
- strongest answers recognise the scale of climate externality (global, long-run, irreversible) demands multiple complementary instruments, and reach a supported judgement

Use the levels mark scheme for 25-mark questions to award candidates marks for this question.

END OF MARK SCHEME